

# MD. ROFAZ HASAN RAFIU

AI Engineer • Generative AI & Computer Vision • Backend Systems

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## PROFESSIONAL SUMMARY

Computer Science & Engineering student at Rajshahi University of Engineering & Technology (RUET) building AI-powered software systems across Generative AI, computer vision, and backend engineering. Experienced in integrating local language models, structured AI workflows, client-side computer vision, REST APIs, PostgreSQL, and real-time systems. Focused on developing reliable AI applications with practical fallbacks, validation, and production-oriented software architecture.

## EDUCATION

**Rajshahi University of Engineering & Technology (RUET) — *B.Sc. in Computer Science & Engineering*** 2023 – 2027  
• Admitted through national combined engineering university admission process (CUET, KUET, RUET).  
• Relevant Coursework: Linear Algebra, Probability & Statistics, Algorithms & Data Structures, Database Systems, Discrete Mathematics.

**St. Joseph Higher Secondary School — *Higher Secondary Certificate (HSC), Science*** 2022  
• GPA: 5.00 / 5.00.

## TECHNICAL SKILLS

**AI & Machine Learning:** Machine Learning, Generative AI, Large Language Models (LLMs), Local SLM Deployment (Ollama, Qwen 2.5), Model Inference, Structured Outputs, Prompt Engineering, Deterministic Fallbacks, AI Evaluation.

**Computer Vision:** OpenCV.js (WebAssembly), TensorFlow.js (BlazeFace, WebGL), Image Preprocessing (Adaptive Thresholding, Canny Edge Detection, 4-Point Homography Warp, Contour Analysis, ROI Segmentation), Pixel Intensity Analysis.

**Programming & Frameworks:** Python (FastAPI, NumPy, OpenCV basics), TypeScript, Go 1.22+, C++, React 19, Next.js 15, Flutter (Dart).

**Backend, Data & Infrastructure:** PostgreSQL (Prisma ORM, Indexing, Connection Pooling), Redis, SQLite, Docker, Linux/Bash, WebSockets, RESTful APIs, Edge CDN Caching, Git, GitHub Actions.

## SELECTED AI & SOFTWARE ENGINEERING PROJECTS

**Digital School — AI Systems Engineer** [Next.js 15, TypeScript, PostgreSQL, Prisma, Ollama, Qwen 2.5, TensorFlow.js] 2024 – Present  
Full-stack education platform combining automated assessment, diagnostic analytics, computer vision, and backend services. Live: <https://rofazacademy.dev>

- **Local SLM Inference:** Integrated Qwen 2.5 via Ollama to classify student exam errors across structured academic concept categories using constrained prompts and structured schema outputs.
- **Deterministic Resilience:** Implemented a PostgreSQL-based deterministic fallback for diagnostic reporting when local model inference is unavailable or saturated, ensuring continuous service uptime.
- **Client-Side Computer Vision:** Developed browser-based computer vision workflows using TensorFlow.js and WebGL for real-time examinee presence tracking without server video streaming overhead.
- **Data Infrastructure & Caching:** Designed multi-tenant PostgreSQL schemas with Prisma ORM and configured edge CDN caching headers, optimizing connection pooling for concurrent assessment workloads.

**OMRView — Computer Vision Engineer** [TypeScript, React 19, Vite, OpenCV.js WebAssembly] 2025  
Client-side optical mark recognition and document geometry rectification engine. Code: [github.com/rofazhasan/OMRView](https://github.com/rofazhasan/OMRView)

- **WebAssembly Processing:** Moved document-processing operations to client-side OpenCV.js/WebAssembly, executing image rectification and mark recognition directly in the browser.
- **Geometric Rectification:** Implemented a multi-stage vision pipeline with adaptive thresholding, Canny edge detection, and 4-point homography perspective warps to rectify skewed camera captures.
- **Mark Classification:** Constructed a bubble fill classifier evaluating pixel intensity histograms across segmented Regions of Interest (ROIs) against baseline illumination to detect filled marks and multi-fill collisions.

**FocusGuard — Systems Software Engineer** [Go 1.22, Docker, WebSockets, Chrome MV3, SQLite] 2025  
Cross-platform daemon and browser extension for real-time digital policy enforcement. Code: [github.com/rofazhasan/FocusGuard](https://github.com/rofazhasan/FocusGuard)

- **Distributed Daemon:** Built a centralized Go 1.22 service coordinating real-time focus session state across desktop and mobile clients using persistent WebSockets.
- **Browser-Level Policy:** Implemented declarative browser-level network filtering rules using Chrome's Declarative Net Request API, avoiding per-request JavaScript interception overhead.
- **Local-First Synchronization:** Engineered local SQLite policy caching with delta synchronization and integrated an Android VPN UDP DNS sinkhole intercepting restricted domains.

**Krishi Bondhu AI — Mobile & Edge Engineer** [Flutter, Dart, Riverpod, SQLite, Camera Hardware] 2024  
Offline-first mobile crop health assistant for field image collection and diagnosis. Code: [github.com/rofazhasan/Krishi\\_Bondhu\\_AI](https://github.com/rofazhasan/Krishi_Bondhu_AI)

- Developed an on-device camera image acquisition pipeline with standardized leaf capture, local SQLite persistence, and bilingual audio feedback to support low-connectivity agricultural field use.

## HONORS & LEADERSHIP

**Creative Talent Hunt — Regional Champion, Mathematics & Computing** Government of Bangladesh • 2018  
Recognized by the Government of Bangladesh for demonstrated excellence in mathematics, analytical reasoning, and computing.

**RUET CSE Peer Workshop Instructor & Competitive Programmer** 2023 – Present  
Organized and taught hands-on peer workshops on Data Structures, Algorithms, and System Architecture; active participant in programming contests.